WiFi CSI-Based Gait / Person Recognition

Q1-Level Architecture, Dataset, Experiment, and Implementation Report

Prepared from the complete research discussion in this chat

Dataset: Intel AX210 receiver, TP-Link Archer C6 v2 AP, PicoScenes AP-mode CSI collection
Regular setup: 25 participants x 100 samples; modified antenna-spread setup: 5 overlapping participants x 100 samples

Generated report deliverables: DOCX and PDF

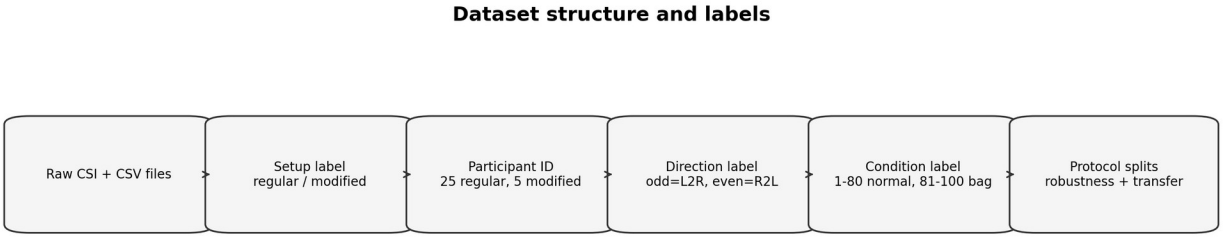

Figure 1. Dataset structure and deterministic label rules used throughout the report.

Important note: All claims about novelty are phrased conservatively. Where literature coverage may be incomplete, the report uses "likely underexplored" rather than "never implemented".

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1. Executive diagnosis and strict Q1 verdict

Your dataset is publishable, but it is not automatically Q1-ready. Its strongest scientific value is not simply "25 people with WiFi CSI gait samples." The stronger story is a WiFi CSI gait/person-recognition dataset with controlled direction labels, carrying-condition variation, and a paired regular-versus-antenna-spread router subset that can support robustness, domain-shift, and representation-learning studies.

The regular-router setup provides 25 participants x 100 samples = 2,500 samples. The modified-router setup provides 5 overlapping participants x 100 samples = 500 samples. Each sample has a CSI file and a CSV file. Odd/even sample indices encode walking direction, and samples 1-80 versus 81-100 encode normal walking versus bag-carrying walking.

The best Q1 route is not a simple accuracy paper. A paper that says "we trained CNN/LSTM/Transformer and achieved high accuracy on 25 people" will look incremental. A Q1-level paper must emphasize hard evaluation protocols, anti-leakage splitting, cross-direction and cross-bag generalization, cross-setup/domain adaptation, few-shot or metric learning, statistical reliability, and explainability.

Table 1. Dataset label axes and research value.

Axis	Labels / values	Scientific use
Person identity	25 regular, 5 modified-overlap	Main biometric/person-recognition target.
Direction	left-to-right / right-to-left	Robustness and equivariance target.
Carrying condition	normal / with bag	Covariate shift and condition invariance target.
Setup/domain	regular router / antenna-spread router	Cross-configuration transfer and domain adaptation target.

The strongest Paper 1 should focus on condition-invariant, direction-equivariant identity representation learning using the 25-person regular-router dataset. The strongest Paper 2 should focus on paired regular-to-modified antenna-configuration domain adaptation using the 5 overlapping participants, with conservative claims unless more modified-router participants are collected.

2. Dataset source, label space, and notation

The dataset description used in this report comes from the uploaded CSI Dataset Structure document and the accompanying pasted research prompt. The technical setup is Intel AX210 as receiver, TP-Link Archer C6 v2 as AP router, PicoScenes CSI tool, and AP-mode data collection.

2.1 Hardware and setup

- Receiver: Intel AX210.
- Access point/router: TP-Link Archer C6 v2.
- CSI tool: PicoScenes CSI tool.
- Regular-router configuration: no physical router modification.
- Modified-router configuration: two 5 GHz router antennas were detached and positioned approximately 4 m apart to create a wider spatial span over the walking path.

2.2 Dataset size and deterministic labels

- Regular-router setup: 25 participants, 100 samples per participant, one CSI file and one CSV file per sample.
- Modified-router setup: 5 participants, 100 samples per participant, and these 5 participants also appear in the regular-router setup.
- Odd-indexed samples: left-to-right walking.
- Even-indexed samples: right-to-left walking.
- Samples 1-80: normal walking.
- Samples 81-100: walking with bag; odd/even direction still applies.

2.3 Mathematical notation

For a mini-batch of CSI windows, define the input tensor as:

$$X \in \mathbb{R}^{B \times T \times S \times L \times C} \quad (1)$$

where B is batch size, T is packet/time length, S is the number of subcarriers or subcarrier groups, L is the number of antenna links or spatial streams, and C is the modality dimension, for example amplitude only or amplitude plus phase.

$$y \in \{1, \ldots, N\}, \quad d \in \{0, 1\}, \quad c \in \{0, 1\}, \quad s \in \{0, 1\} \quad (2)$$

Here, y denotes person identity, d denotes walking direction, c denotes carrying condition, and s denotes setup/domain. In this dataset, d=0 may denote left-to-right and d=1 right-to-left; c=0 normal and c=1 with bag; s=0 regular router and s=1 modified antenna-spread router.

```
participant_id = folder/person identifier

sample_id      = integer parsed from filename


if sample_id % 2 == 1: direction = left_to_right

else:                  direction = right_to_left


if 1 <= sample_id <= 80:   condition = normal

if 81 <= sample_id <= 100: condition = with_bag


setup_type = regular_router or modified_antenna_spread
```

3. Current literature verification and positioning

The field has already moved beyond simple classifiers. Early works used handcrafted features and SVM-style models. Later works used CNNs, CNN-LSTMs, CNN-BiGRUs, ViTs, generic Transformers, self-supervised learning, few-shot learning, GAN/domain adaptation, and cross-domain adaptation. Therefore, a generic CNN-LSTM or Transformer is baseline-level, not a Q1 architecture contribution.

3.1 Key literature comparison

Table 2. Literature comparison used for research positioning

Paper / year	Dataset	Task	Model used	Input	Key contribution	Limitations	Novelty of your direction
WiWho, 2016	20 volunteers; small groups	WiFi gait person ID	Handcrafted gait/CSI features	CSI walking perturbation	Early proof of device-free CSI gait ID	Small group evaluation; limited robustness	Novel if you focus on direction/bag/domain robustness, not just ID.
WiFi-ID, 2016	COTS WiFi; 20-subject style evaluation	Human ID	Time/frequency features + ML	CSI perturbation features	Showed walking-induced CSI can identify people	Classical ML; weak scalability and robustness	Your factorized deep model is novel relative to it; ML baselines still mandatory.
NeuralWave, 2018	Private dataset; 24 people	Gait-based user ID	23-layer CNN / RadioNet	Sanitized CSI / spectrogram-like	Early deep learning for CSI gait biometrics	CNN-only; no nuisance disentanglement	Plain CNN is not novel; RF-axis-aware variants may be.
CSIID, 2019	COTS WiFi	Human identification	CNN + LSTM	CSI time series	Combined convolution and recurrence	CNN-LSTM now overused	Use only as baseline.
Gate-ID, 2020	20 subjects crossing back/forth	Direction-invariant ID	Direction estimation + augmentation/attention	CSI spatial-temporal features	Explicitly studied direction and mirror-like CSI patterns	No bag/no-bag or antenna-spread transfer	Your direction-equivariant + bag + setup design extends the gap.
CAUTION, 2022	WiFi gait authentication	Few-shot open-set authentication	Few-shot profiles + threshold	CSI gait features	Limited-enrollment authentication and intruder rejection	Not focused on direction/bag/setup factorization	Few-shot/open-set can be redesigned around your covariates.
AutoFi, 2022	Atheros + public datasets	Automatic WiFi sensing	Geometric self-supervised learning	Unlabeled CSI samples	Annotation-efficient WiFi sensing	Not identity/direction/bag/setup -specific	Masked/self-supervised CSI pretraining can be strong if CSI-specific.
SenseFi, 2023	Multiple public CSI datasets	WiFi sensing benchmark	MLP, CNN, RNN, Transformer baselines	CSI tensors	Benchmark/library for WiFi sensing	Generic models are baseline-level	Architecture must exploit your labels and hard protocols.
Wi-Gait, 2023	CSI gait/person ID dataset	Robust personnel identification	Multiscale residual convolution-enhanced Transformer	WiFi CSI gait features	Transformer-based robust gait ID	Generic Transformer novelty is reduced	Transformer proposals need factorization, physics, domain, or SSL.
DCS-Gait, 2024	34 subjects, 2 scenes, 3 states	Cross-scene/state gait recognition	Class-level domain adaptation	WiFi CSI	Strong cross-domain CSI gait adaptation	Generic DA already explored	Your novelty must be paired router-configuration adaptation.
MaP-SGAN, 2024	Cross-domain CSI gait	Cross-domain gait recognition	Multi-anchor point Siamese GAN	WiFi CSI	GAN-based asymmetric domain adaptation	GAN/DA already used	Avoid GAN novelty unless physically paired and carefully evaluated.
NeuralWiGait, 2024/2025	Widar 3.0 and NTU-Fi-HumanID	WiFi gait recognition	CNN + BiGRU	CSI samples	Hybrid recurrent deep CSI gait model	CNN-BiGRU now baseline-level	Use as baseline, not contribution.
ViT for WiFi CSI HAR, 2024	UT-HAR, NTU-Fi HAR	Activity recognition	ViT, SimpleViT, DeepViT, Swin, CaiT	CSI spectra/images	ViT family applied to CSI HAR	HAR, not gait ID; ViT-on-CSI image no longer novel	CSI-specific tubelets and covariate learning needed.
WiSigPro, 2024	CSI HAR	Passive WiFi sensing	Transformer with signal processing	CSI spatiotemporal data	Attention-based WiFi sensing	Generic attention saturated	Avoid MHA-only novelty.
Multi-view CSI gait fusion, 2025/2026	CSI gait recognition	Few-sample gait recognition	GADF, heatmaps, ViT, CNN+Transformer	Raw CSI + image views	Multi-view feature fusion	Reduces novelty of simple amplitude/spectrogram fusion	Fusion must add covariate robustness or domain transfer.
Transformer amplitude/phase person ID, 2025	ESP32, 6 participants	Stationary person ID	Dual-branch Transformer	Amplitude + phase	Separate amplitude/phase branches	Stationary and small participant count	Amplitude+phase alone is not enough; use gait-specific losses.
CSI-Bench, 2025	35 users, 26 environments, 16 device types, 460+ hours	Multitask WiFi sensing benchmark	Standardized baselines	Commercial CSI	Raises generalization/reproducibility bar	Much larger than your dataset	Your paper must be honest about single-environment limits.

3.2 Overused directions that should not be main novelty

Overused direction	Reviewer reaction
CNN on CSI images	Too common; should be a baseline.
CNN-LSTM, CNN-GRU, CNN-BiGRU	Already used; baseline-level.
Vanilla Transformer encoder	Already common in WiFi sensing.
ViT on CSI spectrogram	Explored in CSI HAR and multi-view CSI papers.
Softmax-only person classification	Too shallow for a biometric/gait Q1 paper.
Random split only	Leakage concern and weak generalization.
Amplitude + phase fusion without phase sanitization	Technically weak and risky.
GAN augmentation without hard domain protocol	Superficial.
Pure LLM over serialized CSI rows	Gimmicky and physically unjustified.

3.3 Likely underexplored directions for CSI gait recognition

- Latent factorization of identity, direction, bag condition, and router setup in a single CSI gait architecture.
- Direction equivariance: modeling left-to-right and right-to-left walking as a learnable latent transformation rather than as simple data augmentation.
- Paired regular-modified router adaptation using the same five subjects as a cross-setup bridge.
- Antenna-spread physical priors: treating the 4 m antenna-spread setup as a spatial-aperture/domain shift.
- Hypergraph modeling of time x subcarrier x antenna-link CSI structure.
- Masked amplitude-phase CSI pretraining with identity-preserving fine-tuning.
- Counterfactual condition invariance: same identity under normal/bag and opposite direction.
- Open-set biometric verification under direction, bag, and setup shift, not only closed-set classification.

4. Deep dataset analysis and leakage risks

4.1 Strengths

- The 25-person regular-router subset is respectable for WiFi CSI gait/person recognition and is larger than many early CSI gait ID demonstrations.
- The 100 repeated samples per participant allow controlled train/validation/test splits, few-shot simulations, and reliability testing.
- Deterministic direction and bag/no-bag labels enable serious robustness protocols, not just standard classification.
- The modified antenna-spread subset is genuinely interesting because it creates a physical router-configuration domain shift.
- The five overlapping subjects in regular and modified setups enable paired cross-domain adaptation and prototype alignment.

4.2 Weaknesses

- If all data were collected in one room/corridor, environment-specific multipath may dominate identity features.
- The modified-router subset has only five participants, which is weak for a standalone high-impact antenna-setup claim.
- The bag condition is imbalanced: 80 normal samples and 20 bag samples per participant.
- Continuous collection may create session leakage if random sample/window splits are used.

- Important metadata may be missing: room geometry, AP/RX positions, antenna height/orientation, CSI dimensions, packet rate, channel, bandwidth, packet drops, demographics, bag weight, and session conditions.

4.3 Missing metadata that should be documented

Category	Details to record before training
Hardware	Intel AX210 receiver, TP-Link Archer C6 v2 AP, antenna count, orientation, RF-chain mapping.
WiFi PHY	Band, channel, bandwidth, 802.11 mode, MCS, NSS, guard interval, transmit power, traffic generation.
PicoScenes	Version, command, exported fields, CSI dimension, timestamp format.
Geometry	Room size, AP-RX distance, walking path coordinates, line-of-sight/NLOS, antenna heights.
Modified setup	Exact 4 m antenna coordinates, cable type/length, which antennas were detached, cable loss control.
Participant metadata	Anonymized ID, age range, sex, height, weight, shoe type if available.
Bag condition	Bag type, approximate weight, carrying side, consistency of carrying.
Collection process	Sample order, breaks, date/time, background motion, WiFi interference.
File integrity	Packet count, duration, dropped-packet rate, missing/corrupted files.
Ethics	Consent, anonymization, biometric-risk disclosure, intended use.

4.4 Leakage risks and fixes

Leakage risk	Why dangerous	Fix
Window leakage	Multiple windows from the same CSI sample appear in train and test.	Split at sample level before windowing.
Session leakage	Nearly adjacent samples from the same continuous collection appear in both train and test.	Use block/sample-index splits and hard condition splits.
Preprocessing leakage	Normalization/PCA fitted using all data leaks test statistics.	Fit scaler/PCA only on training data.
Augmentation leakage	Augmented version of a test sample appears in training.	Apply augmentation only to training samples.
Setup leakage	Same modified target samples used for tuning and final testing.	Separate adaptation, validation, and locked target test sets.
Hyperparameter leakage	Test set repeatedly used for model selection.	Use validation set; keep final test locked.

4.5 Strongest split suite for Q1

19. In-distribution 25-person regular closed-set identification.
20. Cross-direction identification: L2R to R2L and R2L to L2R.
21. Cross-bag identification: normal walking to with-bag walking.
22. Combined hard split: normal L2R to bag R2L and normal R2L to bag L2R.
23. Few-shot enrollment and open-set verification.
24. Cross-setup transfer: regular to modified and modified to regular on the five overlapping subjects.
25. Ablation and statistical testing across all protocols.

5. Publishable research gaps and improvement scope

Improvement area	Why it matters	Q1 strength	Required experiments
Signal preprocessing	CSI has packet drops, receiver gain changes, and phase errors.	Medium alone; high if linked to robustness.	Raw vs filtered vs calibrated under hard splits.
Amplitude/phase cleaning	Amplitude is stable; phase can	High if phase improves hard	Amplitude-only, phase-only,

	add spatial information but needs sanitization.	splits.	fused with train-only normalization.
Feature representation	Different views capture different gait dynamics.	High if tested under hard protocols.	Raw tensor, PCA, spectrogram, token sequence, statistical vectors.
Architecture novelty	Generic CNN/Transformer is saturated.	High if architecture exploits identity/direction/bag/setup labels.	Compare with CNN, CNN-GRU, TCN, Transformer, ArcFace, SupCon.
Robustness	Real CSI gait systems fail under direction/carrying changes.	Very high.	Cross-direction, normal-to-bag, combined hard shifts.
Domain adaptation	Modified setup is a physical setup/domain shift.	Medium-high with five subjects; higher with more data.	R->M, M->R, DA, few-shot target adaptation.
Few-shot learning	Biometric systems need enrollment with few samples.	High.	5-shot, 10-shot, 20-shot, open-set verification.
Explainability	Reviewer trust improves when subcarrier/time importance is interpretable.	Medium-high.	Saliency, attention, UMAP, antenna-link ablation.
Deployment	CSI sensing papers need practical feasibility.	Medium.	FLOPs, latency, memory, window length, subcarrier reduction.
Statistical reliability	Single-run results are weak.	Mandatory.	Mean +/- std, 95% CI, Wilcoxon/t-test, McNemar.

6. Model families and baseline hierarchy

The strict hierarchy below separates mandatory baselines from actual Q1-level contributions. A serious paper must compare the proposed architecture against classical ML, CNN/recurrent baselines, TCN, Transformer, metric learning, and domain-adaptation baselines where relevant.

Category	Models	Input format	Contribution level
Traditional ML	SVM, RF, XGBoost/LightGBM/CatBoost, KNN, LR, LDA/QDA, HMM	Statistical vectors, PCA, Doppler features	Mandatory baseline.
Classical DL	1D CNN, 2D CNN, CNN-LSTM/GRU, BiLSTM/GRU, TCN, autoencoder	Amplitude tensor, spectrogram, PCA sequence	Baseline to good-level.
Metric learning	Siamese, triplet, prototypical, ArcFace/CosFace/SphereFace	CSI embeddings	Q1-capable if used for few-shot/open-set protocols.
Transformer family	Vanilla Transformer, temporal Transformer, Conformer, ViT, Swin, PatchTST-style	Tokenized CSI, spectrogram patches	Contribution only if CSI-specific and covariate-aware.
Self-supervised	SimCLR, MoCo, BYOL, Barlow Twins, VICReg, TS2Vec, CPC, masked CSI modeling	Unlabeled CSI windows	Q1-capable if paired with few-shot/hard-shift gains.
Domain adaptation	CORAL, MMD, DANN, domain-specific BN, test-time adaptation, meta-learning	Regular/modified CSI embeddings	Q1-capable for Paper 2 but limited by five modified subjects.
LLM-inspired	Masked CSI modeling, autoregressive CSI prediction, codebook Transformer	CSI patches/tokens	Safe only if framed as self-supervised time-series/RF modeling, not pure LLM.

A pure LLM classifier is not scientifically justified for CSI gait recognition. CSI is numerical RF time-series data, not language. Use LLM-inspired ideas only as masked/autoregressive token modeling or explanation/reporting layers after the CSI model output.

7. Type A: completely new architecture ideas

The following architecture ideas are framed conservatively as apparently novel or likely underexplored for CSI-based gait/person recognition. In a manuscript, use "to the best of our knowledge" and support the claim with a careful related-work table.

A1. CID-GaitNet: Condition-Invariant Disentangled CSI Gait Network

Q1 title: Condition-Invariant Disentangled Representation Learning for WiFi CSI Gait Biometrics Under Direction and Carrying Variations

Item	Specification
Core idea	Learn separate latent subspaces for identity, walking direction, and bag condition. The identity subspace is discriminative for person ID but adversarially uninformative about direction and carrying condition.
Novelty	CSI gait works often use augmentation or auxiliary labels, but full identity-direction-bag disentanglement appears underexplored.
Dataset suitability	Best fit for the 25-person regular-router subset because every sample has identity, direction, and condition labels.
Input format	Amplitude or amplitude + sanitized phase CSI tensor X in R^{B x T x S x L x C}.
Pipeline	CSI tensor -> preprocessing -> separable CSI stem -> temporal-subcarrier encoder -> shared embedding -> identity/direction/condition projectors -> ArcFace identity head -> direction and bag heads -> adversarial nuisance heads.
Loss functions	ArcFace + SupCon identity + direction CE + bag CE + adversarial direction/bag losses on identity + orthogonality + consistency.
Training strategy	Balanced mini-batches with same identity across direction and condition where possible; start amplitude-only; add sanitized phase after stability.
Evaluation protocol	Random/grouped, L2R->R2L, R2L->L2R, normal->bag, normal-L2R->bag-R2L, few-shot/open-set.
Required baselines	SVM/RF/XGBoost, CNN, CNN-GRU, TCN, Transformer, Wi-Gait-style Transformer, ArcFace-only, SupCon-only.
Ablation studies	Remove adversarial heads, orthogonality, SupCon, direction head, bag head, amplitude/phase/fusion.
Expected weakness	Bag samples are limited and imbalanced.
Q1 defense	The architecture explicitly factorizes nuisance covariates and validates learned invariance under leave-condition-out protocols.
Paper fit	Paper 1.

A2. CycleDIR-CSI: Direction-Equivariant Cycle Transformer

Q1 title: Learning Direction Equivariance in WiFi CSI Gait Recognition via Latent Cycle Constraints

Item	Specification
Core idea	Model L2R and R2L walking as a learnable latent transformation while identity remains invariant.
Novelty	Instead of treating direction as a label or augmentation, it defines an equivariant latent operator Q with Q^2 approximately I.
Dataset suitability	Odd/even sample rules provide abundant opposite-direction samples per subject.
Input format	Amplitude or amplitude + phase CSI tensor.
Pipeline	CSI encoder -> latent identity and direction-sensitive motion vectors -> learnable direction operator Q -> identity and direction heads -> cycle consistency.
Loss functions	ArcFace, direction CE, SupCon, cycle loss, involution loss ||Q^2-I||, identity consistency.
Training strategy	Sample pseudo-pairs of same person, same condition, opposite direction.
Evaluation protocol	Cross-direction and combined cross-direction + bag protocols.
Required baselines	Gate-ID-style augmentation, CNN-GRU, TCN, Transformer, CID-GaitNet without Q.
Ablation studies	No Q, no cycle loss, no pseudo-pair mining, random vs structured pairs.
Expected weakness	Opposite-direction samples are pseudo-pairs, not exact repeated trajectories.
Q1 defense	Direction is encoded as a latent transformation rather than a categorical nuisance only.
Paper fit	Paper 1.

A3. PAIR-BridgeNet: Paired Antenna-Configuration Invariant Router Bridge Network

Q1 title: Paired Cross-Configuration Domain Adaptation for WiFi CSI Gait Recognition Between Compact and Antenna-Spread Router Setups

Item	Specification
Core idea	Use the five same-subject regular/modified pairs to learn a bridge between compact-router CSI and antenna-spread-router CSI.
Novelty	Domain adaptation exists, but paired router-configuration adaptation using a 4 m antenna-spread setting is a distinct physical domain-shift study.
Dataset suitability	Directly uses the five participants with both regular and modified-router samples.
Input format	Regular and modified CSI tensors, amplitude or amplitude+phase.
Pipeline	Regular stem + modified stem -> shared encoder -> identity embedding -> prototype memory -> domain adversarial head -> pair alignment and identity head.
Loss functions	ArcFace, SupCon, domain adversarial, MMD/CORAL, pair alignment, prototype consistency.
Training strategy	Pretrain on 25 regular subjects, adapt using five overlapping regular/modified subjects.
Evaluation protocol	R->M, M->R, few-shot target adaptation, no-adaptation vs adaptation.
Required baselines	No adaptation, fine-tuning, CORAL, MMD, DANN, domain-specific BN, DCS-Gait-inspired class-level DA.
Ablation studies	No pair loss, no domain loss, no prototype memory, no setup-specific stem, no pretraining.
Expected weakness	Five modified subjects are not enough for broad deployment claims.
Q1 defense	Frame as a paired pilot cross-configuration domain-shift study, not a universal antenna design claim.
Paper fit	Paper 2.

A4. Fresnel-HyperGait: Physics-Guided Hypergraph Transformer

Q1 title: A Physics-Guided Hypergraph Transformer for WiFi CSI Gait Recognition With Antenna-Spread Spatial Priors

Item	Specification
Core idea	Represent CSI as a hypergraph over time patches, subcarrier groups, antenna links, and setup geometry.
Novelty	A CSI-gait-specific hypergraph over RF acquisition axes and antenna-setup geometry appears underexplored.
Dataset suitability	Useful for comparing regular and antenna-spread setups if antenna-link metadata is reliable.
Input format	CSI tensor plus setup and antenna-link metadata tokens.
Pipeline	CSI tensor -> patch embedding -> node construction -> hyperedges for time/subcarrier/link/setup -> hypergraph attention -> identity/direction/bag/setup heads.
Loss functions	ArcFace, SupCon, domain adversarial, graph smoothness, subcarrier-link dropout consistency.
Training strategy	Pretrain on regular data; adapt to modified data with geometry-aware hyperedges.
Evaluation protocol	Person ID, setup shift, antenna-link ablation, hyperedge ablation.
Required baselines	Transformer without graph, GCN/GAT, TCN, Wi-Gait-style Transformer.
Ablation studies	Remove hyperedges type-by-type; remove geometry encoding; random graph vs physics graph.
Expected weakness	Requires exact CSI dimension and antenna-link metadata; physical assumptions may be challenged.
Q1 defense	CSI is not a generic image; it is a structured RF tensor with known acquisition axes.
Paper fit	Paper 2 or advanced extension.

A5. M2CSI-MAE: Masked Multi-Modal CSI Autoencoding Transformer

Q1 title: Masked Multi-Modal CSI Modeling for Few-Shot WiFi Gait-Based Person Recognition

Item	Specification
Core idea	Pretrain by masking packets, subcarriers, antenna links, amplitude, and phase, then reconstruct missing CSI and fine-tune for identity.

Novelty	Masked modeling exists, but CSI-specific structured masking across RF axes and biometric fine-tuning is likely underexplored.
Dataset suitability	Uses all regular and modified CSI as unlabeled data and improves few-shot performance.
Input format	Amplitude + sanitized phase CSI tensor.
Pipeline	CSI tensor -> structured mask -> visible-token encoder -> lightweight decoder -> reconstruct amplitude/phase/derivatives -> ArcFace fine-tuning.
Loss functions	Masked reconstruction, phase derivative loss, ArcFace, SupCon.
Training strategy	Self-supervised pretraining on all CSI; supervised fine-tuning on regular labels.
Evaluation protocol	5-shot, 10-shot, 20-shot, normal->bag, L2R->R2L, R->M stress test.
Required baselines	Supervised Transformer, SimCLR, TS2Vec, AutoFi-inspired SSL, MAE without phase.
Ablation studies	Mask ratio, mask type, amplitude-only, phase-only, no reconstruction, no contrastive fine-tuning.
Expected weakness	May not beat supervised training if the dataset is clean and sufficiently labeled.
Q1 defense	Masking is aligned with CSI acquisition axes and downstream biometric invariance.
Paper fit	Paper 1 or separate SSL paper.

A6. ProtoRouter-Gait: Router-Aware Prototype Memory Network

Q1 title: Router-Aware Prototype Memory for Few-Shot and Cross-Setup WiFi CSI Gait Recognition

Item	Specification
Core idea	Maintain identity prototypes and condition/setup-specific prototype adapters for direction, bag, and router setup.
Novelty	Few-shot CSI authentication exists, but condition-specific prototype memory over identity x direction x bag x setup is likely underexplored.
Dataset suitability	Your repeated samples per person are ideal for 5/10/20-shot enrollment simulations.
Input format	Embedding from CSI CNN/TCN/Transformer encoder.
Pipeline	CSI encoder -> embedding -> prototype memory -> condition/setup adapter -> metric classifier -> open-set threshold.
Loss functions	Prototype loss, ArcFace/CosFace, triplet, open-set margin, condition consistency.
Training strategy	Episodic N-way K-shot training with support/query splits.
Evaluation protocol	Few-shot, open-set, cross-bag, cross-direction, cross-setup.
Required baselines	ProtoNet, MatchingNet, RelationNet, CAUTION-style threshold, Siamese, triplet.
Ablation studies	No condition prototype, no setup prototype, Euclidean vs cosine, fixed vs learnable prototypes.
Expected weakness	Less compelling if only closed-set classification is reported.
Q1 defense	Reframes CSI gait as realistic enrollment-based biometrics rather than closed-set classification.
Paper fit	Paper 1.

A7. CSI-CausalGait: Causal Nuisance Deconfounding Network

Q1 title: Causal Nuisance Deconfounding for Robust WiFi CSI Gait Biometrics

Item	Specification
Core idea	Treat identity as target and direction, bag, and setup as observed nuisance factors; learn identity invariant to interventions on nuisances.
Novelty	Causal-inspired invariance is rare in CSI gait recognition.
Dataset suitability	Your labels naturally define environments: normal L2R, normal R2L, bag L2R, bag R2L, modified setup.
Input format	Amplitude or amplitude+phase CSI.
Pipeline	CSI encoder -> latent factors -> identity predictor -> nuisance predictors -> adversarial deconfounding -> counterfactual consistency.
Loss functions	ArcFace, IRM-like environment risk, adversarial nuisance loss, counterfactual consistency, SupCon.
Training strategy	Leave-one-environment-out and GroupDRO/IRM-style optimization.
Evaluation protocol	Hard environment shift protocols.
Required baselines	ERM, GroupDRO, IRM-style baseline, DANN, CID-GaitNet.
Ablation studies	No causal loss, no adversarial nuisance, no environment grouping.
Expected weakness	Causal claims can be challenged; must be framed as causal-inspired,

	not proof of true causality.
Q1 defense	Use conservative language and demonstrate improvements under held-out nuisance environments.
Paper fit	Paper 1 or theoretical extension.

A8. Aperture-MoE: Antenna-Aperture Mixture-of-Experts Network

Q1 title: Antenna-Aperture-Aware Mixture-of-Experts for Cross-Configuration WiFi CSI Gait Recognition

Item	Specification
Core idea	Use setup-aware experts for compact router, antenna-spread router, shared identity, and direction/bag robustness, selected by a gating network.
Novelty	MoE is common elsewhere, but router-configuration-aware MoE for CSI gait is likely underexplored.
Dataset suitability	Useful for regular vs modified setup if experts are small and heavily regularized.
Input format	CSI tensor with setup label during training or inferred setup at inference.
Pipeline	CSI stem -> shared encoder -> setup-aware gating -> expert bank -> fused embedding -> identity/domain/condition heads.
Loss functions	ArcFace, load-balancing, domain loss, condition loss, consistency loss.
Training strategy	Pretrain shared expert on regular data; train setup experts using the five paired modified subjects.
Evaluation protocol	R->M, M->R, mixed-domain ID, expert utilization.
Required baselines	Single encoder, domain-specific BN, DANN, PAIR-BridgeNet.
Ablation studies	Number of experts, hard vs soft gating, no load-balancing, no shared expert.
Expected weakness	500 modified samples may be too few for many experts.
Q1 defense	Use a small expert bank and prove parameter efficiency.
Paper fit	Paper 2.

8. Type B: modified existing architectures with publishable redesigns

Table 3. Modified architecture ideas and Q1 contribution statements

Original	Modified name	Exact modification	Why not trivial	Dataset support	Novelty	Difficulty	Contribution statement
CNN	Subcarrier-Separable CSI-ResCNN	Depthwise/separable convolutions over time, subcarrier, and antenna-link axes with link attention.	Treats CSI axes separately, not as generic image channels.	All 25 regular subjects.	Medium	Easy	RF-axis-aware CNN for CSI gait recognition.
CNN-LSTM/GRU	GaitCycle ConvGRU	Conv encoder plus GRU with gait-cycle attention and condition-balanced sampling.	Uses walking-cycle structure rather than generic recurrent pooling.	Direction and bag labels.	Medium	Medium	Condition-aware temporal modeling of CSI gait cycles.
TCN	Direction-Equivariant TCN	Dilated TCN with latent direction-reversal consistency.	Adds explicit L2R/R2L equivariance.	Odd/even direction labels.	High	Medium	TCN that models walking direction as a latent transformation.
ResNet/Inception	Multi-Scale CSI-InceptionNet	Parallel temporal kernels for slow torso and fast limb motion plus subcarrier branch.	CSI-specific temporal scale design.	Full regular dataset.	Medium	Easy-medium	Multi-scale RF gait feature extraction.
Siamese	Covariate-Paired Siamese CSI-Net	Positive pairs cross direction/bag; negatives share condition but differ by identity.	Pair mining explicitly targets invariance.	Full regular dataset.	High	Medium	Covariate-aware Siamese learning for CSI biometrics.
Triplet	Hard-Covariate Triplet GaitNet	Anchor normal-L2R, positive bag-R2L same person, negative same condition different person.	Triplets enforce hardest real shifts.	Direction + bag labels.	High	Medium	Hard-condition triplet mining for device-free gait biometrics.
Vision Transformer	CSI Tubelet ViT	3D tubelets over time x subcarrier x antenna-link instead of 2D CSI image patches.	Preserves CSI tensor structure.	Amplitude/spectrogram inputs.	Medium-high	Medium	CSI tensor tubelet tokenization for gait.
Time-Series Transformer	PatchTST-CSI	Patch time series per subcarrier/link with shared backbone and cross-channel fusion.	Adapts patch time-series idea to CSI axes.	All CSI tensors.	Medium-high	Medium	Patch-based CSI sequence modeling for gait.
Conformer	CSI-Conformer-X	Local convolution for gait micro-dynamics plus attention for long-range walking sequence and subcarrier attention.	Better inductive bias than vanilla Transformer.	25 regular subjects.	Q1-potential	Medium-hard	Conformer redesigned for CSI gait temporal-local structure.
Swin Transformer	Shifted CSI-Window Transformer	Shifted windows over time-subcarrier maps with antenna-link tokens.	Local-global hierarchy for CSI maps.	Spectrogram/CSI image.	Medium	Hard	Hierarchical local attention for CSI gait maps.
GNN	Antenna-Subcarrier GaitGNN	Graph nodes are subcarrier groups and antenna links; temporal edges across packets.	CSI topology encoded explicitly.	Stable subcarrier/link metadata.	High	Hard	Graph-structured RF representation learning for gait.
Autoencoder	Disentangled CSI Denoising AE	Separate identity/nuisance latent variables; reconstruct CSI while adversarially removing nuisance.	Reconstruction plus invariance, not simple denoising.	All data.	Medium-high	Medium	Nuisance-aware CSI reconstruction for robust biometrics.
Masked AE	Condition-Aware CSI-MAE	Structured masking by packet, subcarrier, antenna link, amplitude/phase, followed by ArcFace fine-tuning.	CSI-specific masking strategy.	All regular + modified.	Q1-potential	Medium-hard	Masked CSI modeling for few-shot gait recognition.
Contrastive	Multi-Condition SupCon CSI	Positives are same identity across direction/bag/setup; hard negatives share condition.	Uses labels deeply.	Full regular + 5 cross-setup.	Q1-potential	Medium	Condition-aware contrastive learning for CSI gait.
Domain adaptation	Paired Router DANN++	DANN plus paired prototype alignment and setup-specific stems.	Uses paired same-subject domains.	Five overlap subjects.	Q1-potential but data-limited	Hard	Paired router-configuration domain adaptation.
Multi-task learning	Identity-Nuisance MTLNet	Joint ID/direction/bag/setup heads with gradient reversal from identity branch.	MTL plus deconfounding.	All labels.	High	Medium	Multi-task deconfounded CSI gait learning.
Lightweight model	EdgeCSI-GaitLite	Mobile depthwise CNN plus tiny TCN plus ArcFace plus quantization-aware training.	Deployment-aware, not only accuracy.	All data.	Medium	Medium	Robust edge-deployable CSI gait model.

9. LLM-inspired architecture analysis

Question	Strict answer
Is a pure LLM suitable for CSI gait recognition?	No. CSI is numerical RF time-series data, not language. A pure text LLM classifier would look gimmicky.
Is it valid to tokenize CSI like language?	Yes, if tokens are meaningful CSI patches: time patches, subcarrier groups, antenna links, amplitude/phase modalities. Do not call this an LLM unless it truly uses language modeling.
Can Transformer-token modeling be used without calling it an LLM?	Yes. Use terms such as tokenized CSI sequence modeling, masked CSI modeling, or RF time-series Transformer.
Can masked CSI modeling be inspired by BERT/MAE?	Yes. This is one of the safest LLM-inspired ideas.
Can GPT-style autoregressive prediction be useful?	Yes as self-supervised future-patch prediction, not as a final text-style LLM classifier.
Can an LLM be an explanation layer?	Yes, after the CSI model outputs identity/confidence/saliency/domain-shift scores, an LLM can generate reports. This is not the scientific core.
Would reviewers consider LLM usage gimmicky?	Yes, if CSI is serialized into text or if the model is marketed as LLM-based without RF justification.

9.1 Serious LLM-inspired architectures

Architecture	Serious idea	Why acceptable
CSI-BERT-MAE	Mask CSI patches and reconstruct amplitude, phase, subcarrier, and temporal features; fine-tune for identity.	Grounded self-supervised RF representation learning.
CSI-GPT-Predictor	Autoregressively predict future CSI patches and use encoder states for gait ID.	Learns walking temporal dynamics; use only as pretraining.
CSI Codebook Transformer	Learn a vector-quantized CSI codebook and model discrete RF tokens.	Serious if codebook tokens are RF patches, not text.

9.2 Risky LLM ideas to avoid

- Converting CSI CSV rows into text and fine-tuning an LLM.
- Asking ChatGPT or a closed LLM to classify people from serialized CSI.
- Claiming "LLM-based CSI recognition" because the model has a Transformer encoder.

10. Best two-paper Q1 strategy

10.1 Paper 1: safest Q1 paper

Recommended architecture: CID-GaitNet + CycleDIR.

Tentative title: Condition-Invariant and Direction-Equivariant Representation Learning for WiFi CSI Gait Biometrics.

Central novelty: the model learns identity representations that are invariant to walking direction and bag condition while separately modeling direction and carrying-state factors. The direction module treats L2R and R2L as a learnable equivariant transformation, not merely as an auxiliary label.

Component	Paper 1 design
Dataset	25-person regular-router subset.
Main task	25-class person identification.
Auxiliary tasks	Direction classification and bag/no-bag classification.
Hard protocols	L2R->R2L, R2L->L2R, normal->bag, normal-L2R->bag-R2L.

Minimum baselines	SVM/RF/XGBoost, CNN, CNN-GRU, TCN, Transformer, ArcFace, SupCon, Gate-ID-style direction augmentation.
Ablations	Remove direction head, bag head, adversarial heads, orthogonality, SupCon, ArcFace, CycleDIR, amplitude/phase/fusion.
Journals	IEEE Internet of Things Journal, IEEE Sensors Journal, IEEE TIM, Computer Networks, Expert Systems with Applications.

10.2 Paper 2: different and domain-adaptive

Recommended architecture: PAIR-BridgeNet.

Tentative title: Paired Cross-Configuration Domain Adaptation for WiFi CSI Gait Recognition Between Compact and Antenna-Spread Router Setups.

The five overlapping subjects are useful for a paired cross-configuration pilot, not for universal antenna-design claims. The paper must frame regular-to-modified and modified-to-regular transfer conservatively.

Component	Paper 2 design
Dataset	Five participants with both regular and modified-router versions; optionally pretrain on all 25 regular participants.
Main task	Regular->modified and modified->regular identity transfer.
Domain shift	Compact router versus 4 m antenna-spread router configuration.
Adaptation strategies	No adaptation, fine-tuning, CORAL, MMD, DANN, domain-specific BN, PAIR-BridgeNet.
Few-shot target	k = 1, 5, 10, 20 target samples per person.
Ablations	Remove pair alignment, domain adversarial loss, MMD/CORAL, prototype memory, setup-specific stems.
Journals	Computer Networks, Ad Hoc Networks, IEEE Sensors Journal, IEEE IoT Journal.

11. Detailed architecture blueprints for top models

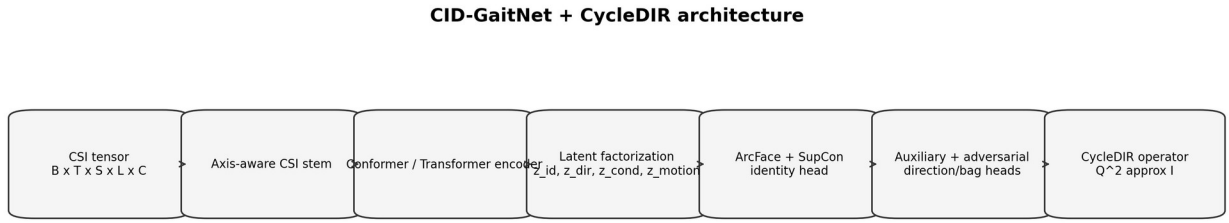

Figure 2. CID-GaitNet + CycleDIR model schematic.

11.1 Blueprint 1: CID-GaitNet + CycleDIR

Let E_theta be the CSI encoder. The encoder maps CSI windows into latent factors:

$$h = E_\theta(X), \quad z_{id} = P_{id}(h), \quad z_{dir} = P_{dir}(h), \quad z_{cond} = P_{cond}(h), \quad z_m = P_m(h) \qquad (3)$$

The identity branch uses an angular-margin biometric head such as ArcFace:

$$\text{logit}_y = s\cos(\theta_y + m), \quad \text{logit}_j = s\cos(\theta_j), \quad j \neq y \quad (4)$$

The supervised contrastive loss for identity embeddings is:

$$L_{\text{SupCon}} = \sum_i \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\text{sim}(z_i, z_p)/\tau)}{\sum_{a \neq i} \exp(\text{sim}(z_i, z_a)/\tau)} \quad (5)$$

Latent factor separation can be encouraged by an orthogonality penalty:

$$L_{orth} = ||Z_{id}^T Z_{dir}||_F^2 + ||Z_{id}^T Z_{cond}||_F^2 + ||Z_{dir}^T Z_{cond}||_F^2 \quad (6)$$

Direction equivariance is modeled with a learnable transformation Q:

$$z_m^{R2L} \approx Q z_m^{L2R}, \quad Q^2 \approx I \quad (7)$$

$$L_{cycle} = ||z_m^{(j)} - Q z_m^{(i)}||_2^2, \quad L_{invol} = ||Q^2 - I||_F^2 \quad (8)$$

Total training objective:

$$L = L_{ArcFace} + \lambda_1 L_{SupCon} + \lambda_2 L_{dir} + \lambda_3 L_{cond} + \lambda_4 L_{adv-dir} + \lambda_5 L_{adv-cond} + \lambda_6 L_{orth} + \lambda_7 L_{cycle} + \lambda_8 L_{invol} \quad (9)$$

Block	Input shape	Output shape	Description
Input	[B,T,S,L,C]	[B,T,S,L,C]	Amplitude or amplitude+phase CSI.
CSI axis stem	[B,T,S,L,C]	[B,N,D]	3D/separable convolution over time, subcarrier, antenna-link, modality axes.
Encoder	[B,N,D]	[B,N,D]	Conformer/Transformer blocks.
Pooling	[B,N,D]	[B,2D]	Attentive mean and standard deviation pooling.
Projectors	[B,2D]	z_id [B,256], z_dir [B,64], z_cond [B,64], z_m [B,128]	Latent factor branches.
Heads	Latents	Logits	ArcFace ID, direction CE, condition CE, adversarial heads.

```
class CIDGaitNet(nn.Module):

  def forward(self, x, person_id=None):

    tokens = self.stem(x)              # [B,N,D]

    tokens = self.encoder(tokens)        # [B,N,D]

    h = self.pool(tokens)              # [B,2D]

    z_id = self.id_projector(h)          # [B,256]

    z_dir = self.dir_projector(h)        # [B,64]
```

```
z_cond = self.cond_projector(h)       # [B,64]

z_motion = self.motion_projector(h)    # [B,128]

id_logits = self.arcface(z_id, person_id)

dir_logits = self.dir_head(z_dir)

cond_logits = self.cond_head(z_cond)

adv_dir_logits = self.dir_adv(grad_reverse(z_id))

adv_cond_logits = self.cond_adv(grad_reverse(z_id))

return {...}
```

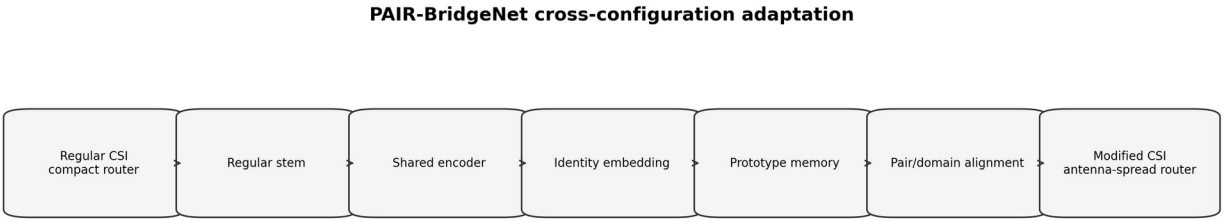

Figure 3. PAIR-BridgeNet model schematic for regular-to-modified router adaptation.

11.2 Blueprint 2: PAIR-BridgeNet

$$z_i^R = E_{shared}(E_R(X_i^R)), \quad z_i^M = E_{shared}(E_M(X_i^M)) \quad (10)$$

$$L_{pair} = ||z_i^R - z_i^M||_2^2 \quad (11)$$

$$p_y^R = \text{mean}(z^R|y), \quad p_y^M = \text{mean}(z^M|y), \quad L_{proto} = \sum_y ||p_y^R - p_y^M||_2^2 \quad (12)$$

$$L = L_{ArcFace} + \lambda_1 L_{pair} + \lambda_2 L_{proto} + \lambda_3 L_{domain} + \lambda_4 L_{MMD/CORAL} + \lambda_5 L_{SupCon} \quad (13)$$

Component	Description
Regular stem	Setup-specific low-level encoder for compact-router CSI.
Modified stem	Setup-specific low-level encoder for antenna-spread CSI.
Shared encoder	Common identity representation learner.
Domain head	Gradient reversal classifier for setup confusion.
Prototype memory	Per-identity prototypes for regular and modified domains.
Pair alignment	Same-subject regular/modified alignment term.

```
z_r = model.encode_regular(x_regular)

z_m = model.encode_modified(x_modified)
```

```
id_loss = arcface_loss(z_r, y) + arcface_loss(z_m, y)

domain_loss = ce(domain_head(grad_reverse(cat([z_r, z_m]))), setup_labels)

loss = id_loss + pair_loss(z_r, z_m) + proto_loss + mmd_loss + domain_loss
```

11.3 Blueprint 3: M2CSI-MAE

$$M \in \{0,1\}^{T \times S \times L \times C}, \quad X_{visible} = X \odot M, \quad X_{masked} = X \odot (1 - M) \quad (14)$$

$$\hat{X} = D(E(X_{visible}), M), \quad L_{rec} = ||(\hat{X} - X) \odot (1 - M)||_1 \quad (15)$$

After pretraining, discard or freeze the decoder and fine-tune the encoder with ArcFace and supervised contrastive losses. Structured masks should cover time packets, subcarrier groups, antenna links, and modality channels.

Mask type	Purpose
Time masking	Forces learning of walking dynamics.
Subcarrier masking	Forces frequency-domain redundancy modeling.
Antenna-link masking	Improves spatial robustness and missing-link tolerance.
Amplitude/phase masking	Encourages cross-modality reconstruction.
Block masking	Prevents trivial interpolation from adjacent packets only.

```
mask = structured_csi_mask(x)

visible = x * mask

latent = encoder(patch_embed(visible, mask))

x_hat = decoder(latent, mask)

pretrain_loss = reconstruction_loss(x_hat, x, mask)


# fine-tune

z = encoder_no_decoder(x)

loss = arcface_loss(z, y) + supcon_loss(z, y)
```

11.4 Blueprint 4: Fresnel-HyperGait

$$G = (V, E), \quad v = \text{token}(t, s, l) \quad (16)$$

Hyperedges can connect tokens sharing time patches, neighboring subcarrier bands, the same antenna link, or the same setup geometry. Hypergraph attention updates token v by aggregating features from hyperedges that contain v:

$$h'_v = \sum_{e \ni v} \alpha_{v,e} W_e \text{aggregate}(\{h_u : u \in e\}) \quad (17)$$

This model is scientifically attractive because CSI is naturally a structured RF tensor rather than a generic image. Its weakness is the requirement for accurate CSI dimension and antenna-link metadata.

Hyperedge type	Physical meaning
Time hyperedge	Packets/patches in the same walking phase.
Subcarrier hyperedge	Neighboring frequency components.
Antenna-link hyperedge	RF-chain/spatial stream relation.
Setup hyperedge	Regular versus antenna-spread spatial geometry.

11.5 Blueprint 5: ProtoRouter-Gait

$$p_y = \text{mean}(\{z_i : y_i = y\}) \quad (18)$$

$$p_y^{d,c,s} = A_{d,c,s}(p_y) \quad (19)$$

$$P(y|x) = \text{softmax}(-\text{dist}(z_x, p_y^{d,c,s})) \quad (20)$$

$$\text{unknown if } \max_y \cos(z_x, p_y) < \tau \quad (21)$$

This architecture is best for realistic biometric enrollment. It should be evaluated with 5-shot, 10-shot, and 20-shot support sets and unknown-person queries.

Module	Role
CSI encoder	Maps each support/query sample to an embedding.
Prototype memory	Stores identity prototypes.
Condition adapter	Maps direction/bag/setup-specific prototypes to canonical identity space.
Metric classifier	Cosine or Euclidean nearest-prototype classification.
Open-set threshold	Rejects unknown persons based on prototype similarity.

12. Novelty and publishability scoring

Table 4. Reviewer-style scoring of architecture ideas

Rank	Architecture	Type	Novelty	Depth	Dataset fit	Feasibility	Risk	Q1 score	Verdict
1	CID-GaitNet + CycleDIR	New	9	9	10	8	4	9	Implement first
2	PAIR-BridgeNet	New	9	9	7	7	7	8	Best Paper 2, data-limited
3	M2CSI-MAE	New/SSL	8	9	9	8	5	8.5	Strong if few-shot gains appear
4	ProtoRouter-Gait	New	8	8	9	8	5	8	Strong biometric framing
5	Fresnel-HyperGait	New	9	9	7	6	8	7.5	Novel but harder
6	CSI-Conformer-X	Modified	7	8	9	8	4	7.5	Strong backbone
7	Hard-Covariate Triplet GaitNet	Modified	7	8	10	8	4	7.5	Good open-set model
8	Multi-Condition SupCon CSI	Modified	7	8	10	9	3	7.5	Excellent practical choice
9	CSI-CausalGait	New	9	9	8	6	8	7	Strong but risky
10	Aperture-MoE	New	8	8	6	7	7	7	Save for Paper 2
11	PatchTST-CSI	Modified	6	7	8	8	4	6.5	Good baseline-plus
12	Antenna-Subcarrier GaitGNN	Modified	8	8	7	6	7	6.5	Needs metadata
13	Condition-Aware CSI-MAE	Modified	7	8	9	8	5	7.5	Close to M2CSI-MAE
14	Direction-Equivariant TCN	Modified	7	7	10	8	4	7	Good ablation/backbone
15	Covariate-Paired Siamese CSI-Net	Modified	6	7	9	8	4	6.5	Good verification baseline
16	CSI Tubelet ViT	Modified	6	7	8	7	5	6	Needs more than tokenization
17	Subcarrier-Separable CSI-ResCNN	Modified	5	6	9	9	3	6	Strong baseline, not main novelty
18	GaitCycle ConvGRU	Modified	5	6	9	8	3	5.5	Baseline-level
19	Multi-Scale CSI-InceptionNet	Modified	5	6	8	9	3	5.5	Good baseline
20	Shifted CSI-Window Transformer	Modified	6	7	7	6	6	5.5	Risk of looking like Swin copy
21	EdgeCSI-GaitLite	Modified	5	6	8	9	3	5.5	Good secondary contribution
22	CSI-GPT-Predictor	LLM-inspired	6	7	7	6	7	5.5	Use only as SSL auxiliary
23	CSI Codebook Transformer	LLM-inspired	7	8	7	5	8	5.5	Interesting but risky
24	Pure LLM CSI classifier	LLM	2	2	2	4	10	1	Avoid

12.1 Strict verdict categories

- Too weak: plain CNN-LSTM, plain Transformer, plain ViT on spectrogram, simple attention CNN, softmax-only classification.
- Too risky: pure LLM classifier, large MoE with only 500 modified samples, strong causal claims without careful protocols, diffusion/GAN counterfactual generation as the main novelty.
- Publishable but need more data: Aperture-MoE, Fresnel-HyperGait, standalone antenna-configuration paper, cross-setup open-set authentication.
- Realistic now: CID-GaitNet + CycleDIR, M2CSI-MAE, Multi-Condition SupCon, ProtoRouter-Gait, Hard-Covariate Triplet, CSI-Conformer-X.

13. Mandatory experimental protocol

Protocol	Train	Test	Purpose
Closed-set ID	Regular-router 25 participants, grouped split	Held-out regular samples	Basic identity recognition without leakage.
Cross-direction D1	L2R only	R2L only	Direction robustness.
Cross-direction D2	R2L only	L2R only	Symmetry check.
Cross-bag B1	Samples 1-80 normal	Samples 81-100 with bag	Carrying-condition robustness.
Hard combined H1	Normal L2R	Bag R2L	Direction + bag shift.
Hard combined H2	Normal R2L	Bag L2R	Reverse hard shift.
Multi-task	Mixed regular samples	Held-out samples	Person + direction + condition recognition.
Few-shot	5/10/20 samples per person	Remaining samples	Enrollment efficiency.
Open-set	20 known persons	5 unknown persons	Unknown-person rejection.
Cross-setup R->M	Regular five overlap subjects	Modified five overlap subjects	Setup transfer.
Cross-setup M->R	Modified five overlap subjects	Regular five overlap subjects	Reverse setup transfer.

13.1 Metrics and statistics

- Closed-set metrics: accuracy, macro-F1, balanced accuracy, per-person F1, confusion matrix.
- Verification/open-set metrics: EER, AUROC, TAR@FAR=1%, TAR@FAR=5%, FAR/FRR curves.
- Robustness metrics: absolute accuracy drop and relative degradation from in-distribution split.
- Statistical reporting: 5 or 10 seeds, mean +/- standard deviation, 95% confidence interval, paired t-test or Wilcoxon signed-rank test, McNemar test for paired classification comparisons.
- Complexity: parameter count, FLOPs, inference latency, GPU/CPU memory, window length sensitivity, subcarrier reduction sensitivity.

14. Dataset preparation and reproducibility pipeline

14.1 Recommended folder organization

```
csi_gait_project/

 data/

  raw/regular/P001/{csi,csv}/

  raw/modified/P001/{csi,csv}/

  processed/{amplitude,phase_sanitized,amp_phase_fusion,spectrogram,pca_features}/

  metadata/metadata_master.csv

  splits/
```

```
scripts/
  01_build_metadata.py
  02_validate_files.py
  03_extract_csi.py
  04_preprocess_csi.py
  05_make_splits.py
  06_train_baselines.py
  07_train_deep.py
  08_eval_protocols.py
experiments/{configs,logs,checkpoints,tensorboard,wandb}/
results/{tables,figures,confusion_matrices,embeddings,statistical_tests}/
manuscript/{paper1,paper2}/
```

14.2 Metadata table design

Minimum metadata columns:

participant_id

sample_id

setup_type

direction

condition

csi_file_path

csv_file_path

collection_date

session_id

router_config

antenna_config

receiver_config

ap_receiver_distance_m

walking_path_length_m

sampling_rate_hz

num_packets

num_subcarriers

num_tx

num_rx

band

channel

bandwidth_mhz

mcs

nss

packet_loss_rate

duration_s

environment_id

bag_weight_kg

notes

checksum

14.3 Preprocessing sequence

26. Load PicoScenes CSI and associated CSV metadata.
27. Validate timestamps, packet counts, CSI dimensions, and file pairing.
28. Remove corrupted packets and abnormal amplitude/RSSI outliers.
29. Extract complex CSI, amplitude, and phase.
30. Apply amplitude denoising: Hampel filter, median filter, low-pass filter, or wavelet denoising.
31. Apply phase unwrapping and sanitization before using phase as model input.
32. Optionally remove static component and apply PCA or Doppler/STFT transformation.
33. Normalize using training-set statistics only.
34. Segment into fixed windows after sample-level split assignment.
35. Apply augmentation only inside training pipeline.
36. Save processed tensors with immutable metadata and checksums.

14.4 Model-ready formats

Format	Shape idea	Suitable models
Raw complex CSI	T x S x L x 2	Advanced fusion models.
Amplitude tensor	T x S x L	CNN, TCN, Transformer, Conformer.
Phase tensor	T x S x L	Phase-specific models after sanitization.
Amplitude + phase	T x S x L x 2	Fusion and MAE models.
Spectrogram	frequency x time x channels	2D CNN, ViT, Swin.
PCA features	T x components	ML, LSTM, TCN.
Statistical vector	feature vector	SVM/RF/XGBoost.
Token sequence	tokens x embedding	Transformer/Conformer/MAE.

15. Manuscript and reviewer-proofing guidance

15.1 Paper title options

- Direction- and Carrying-Condition-Robust WiFi CSI Gait Recognition via Multi-Task Contrastive Learning.

- Condition-Invariant and Direction-Equivariant Representation Learning for WiFi CSI Gait Biometrics.
- Leakage-Safe Evaluation and Robust Representation Learning for WiFi CSI Gait Biometrics.
- Paired Cross-Configuration Domain Adaptation for WiFi CSI Gait Recognition Between Compact and Antenna-Spread Router Setups.

15.2 Manuscript structure

37. Introduction and motivation.
38. Related work: CSI gait/person ID, direction robustness, domain adaptation, self-supervised CSI learning, biometric metric learning.
39. Dataset and experimental setup: hardware, room geometry, labels, metadata, ethics.
40. Methodology: preprocessing, model architecture, losses, training strategy.
41. Experimental protocols: anti-leakage splits and hard generalization tests.
42. Results: baselines, proposed model, robustness, few-shot/open-set, cross-setup.
43. Ablation study: losses, branches, inputs, augmentation, setup adaptation.
44. Explainability: saliency, attention, subcarrier/time importance, embeddings.
45. Complexity and deployment analysis.
46. Limitations and reproducibility.
47. Conclusion.

15.3 Claims to avoid

- Do not claim a universal gait biometric system.
- Do not claim real-world robustness without multiple environments.
- Do not claim privacy-preserving authentication without nuance.
- Do not claim the antenna-spread router solves CSI gait recognition.
- Do not claim LLM-based CSI recognition unless a real, justified language-model component exists.
- Do not claim phase improves performance unless phase sanitization and ablations prove it.

15.4 Weaknesses that may cause rejection

- Single environment without enough metadata.
- Only five modified-router subjects for standalone setup claims.
- Random split only or window leakage.
- No strong baselines.
- No statistical testing.
- No biometric verification metrics.
- No reproducible split files or code/config release plan.
- Overstated deployment or privacy/security claims.

16. Complete PyTorch implementation plan for CID-GaitNet + CycleDIR

16.1 Project structure

```
csi_gait/

  configs/{cid_gaitnet.yaml,baselines.yaml,pair_bridge.yaml}

  csi/readers/picoscenes_reader.py
```

csi/preprocessing/{amplitude.py,phase_sanitization.py,filters.py,windowing.py,normalization.py}

datasets/{csi_dataset.py,episodic_dataset.py,split_protocols.py}

models/{stems.py,encoders.py,pooling.py,heads.py,losses.py,cid_gaitnet.py,pair_bridge_net.py,mae.py}

train/{train_cid.py,train_baselines.py,train_mae.py,train_pair_bridge.py}

eval/{evaluate_closed_set.py,evaluate_cross_direction.py,evaluate_cross_bag.py,evaluate_fewshot.py,evaluate_openset.py, evaluate_cross_setup.py}

logs/{tensorboard,wandb,csv}

results/{tables,figures,checkpoints}

16.2 Dataset item format

```
{
  "x": Tensor[T, S, L, C],
  "person_id": LongTensor[],
  "direction": LongTensor[],   # 0=L2R, 1=R2L
  "condition": LongTensor[],   # 0=normal, 1=bag
  "setup": LongTensor[],       # 0=regular, 1=modified
  "sample_id": int,
  "participant_id": str
}
```

Batch shape:

x: [B, T, S, L, C]

person_id, direction, condition, setup: [B]

16.3 Main class names and tensor shapes

Class / module	Input	Output	Purpose
CSIAxisStem	[B,T,S,L,C]	[B,N,D]	Axis-aware convolution and patch embedding.
CSIConformerBlock	[B,N,D]	[B,N,D]	Local convolution + global attention.
AttentiveStatsPool	[B,N,D]	[B,2D]	Weighted mean and standard deviation pooling.
ProjectionHead	[B,2D]	[B,E]	MLP branch projector.
ArcFaceHead	[B,E], labels	[B,num_classes]	Angular-margin identity classifier.
GradientReverseLayer	embedding	embedding	Adversarial nuisance removal.
DirectionCycleOperator	[B,E_m]	[B,E_m]	Learnable Q with Q^2 approx I.
CIDGaitNet	[B,T,S,L,C]	dict of embeddings/logits	Full architecture.
CIDGaitLoss	outputs + labels	scalar loss	Weighted objective.

16.4 Module skeleton

```
class CSIAxisStem(nn.Module):
```

```
    """Input [B,T,S,L,C]; output [B,N,D]."""


class CSIConformerEncoder(nn.Module):
    """Input [B,N,D]; output [B,N,D]."""


class AttentiveStatsPool(nn.Module):
    """Input [B,N,D]; output [B,2D]."""


class ArcFaceHead(nn.Module):
    """Input embedding [B,E] and labels [B]; output logits [B,num_classes]."""


class DirectionCycleOperator(nn.Module):
    """Learn Q in R^{E_m x E_m}; penalty ||Q^2-I||_F^2."""


class CIDGaitNet(nn.Module):
    def __init__(self, num_identities=25, input_modalities=1,
                 embed_dim=128, id_dim=256, nuisance_dim=64,
                 motion_dim=128, depth=4, heads=4):
        ...


    def forward(self, x, person_id=None, grl_lambda=1.0):
        ...
```

16.5 Recommended loss weights

```
loss:
  arcface: 1.0
  supcon: 0.2
  direction_ce: 0.2
  condition_ce: 0.2
  adversarial_direction: 0.1
  adversarial_condition: 0.1
  orthogonality: 0.05
  cycle: 0.1
```

involution: 0.01

16.6 Training schedule

optimizer: AdamW

learning_rate: 3e-4

weight_decay: 1e-4

batch_size: 32

epochs: 150

warmup_epochs: 10

scheduler: cosine

gradient_clip: 1.0

dropout: 0.2

seeds: [1,2,3,4,5]

early_stopping_metric: hard_split_macro_f1

16.7 Training stages

48. Train ArcFace encoder only on amplitude CSI.
49. Add direction and bag auxiliary heads.
50. Add adversarial nuisance heads with gradient reversal.
51. Add CycleDIR direction operator and cycle/involution losses.
52. Add sanitized phase branch only after amplitude-only pipeline is stable.
53. Optionally add M2CSI-MAE pretraining and compare with training from scratch.

16.8 Experiment logging structure

run_id

git_commit

config_hash

seed

split_name

model_name

input_type

window_length

normalization_method

accuracy

macro_f1

balanced_accuracy

eer

auroc

params

flops

latency_ms

best_epoch

confusion_matrix_path

embedding_umap_path

checkpoint_path

17. Final implementation order and Q1 publication strategy

17.1 Top 10 architecture ideas

54. CID-GaitNet + CycleDIR
55. PAIR-BridgeNet
56. M2CSI-MAE
57. ProtoRouter-Gait
58. Fresnel-HyperGait
59. CSI-Conformer-X
60. Hard-Covariate Triplet GaitNet
61. Multi-Condition SupCon CSI
62. CSI-CausalGait
63. Aperture-MoE

17.2 Exact first experiment to run

Run amplitude-only 25-person regular-router closed-set identification with grouped split, L2R->R2L, and normal->bag protocols. Models: SVM on statistical/PCA features, 1D CNN, TCN, and ArcFace CSI encoder. This verifies whether the dataset has a stable identity signal before complex architecture work.

17.3 Minimum baselines

- SVM
- Random Forest or XGBoost
- 1D CNN
- 2D CNN on CSI image/spectrogram
- CNN-GRU or CNN-BiGRU
- TCN
- Vanilla Transformer
- CSI-Conformer baseline
- ArcFace encoder
- SupCon encoder
- DANN/CORAL/MMD for cross-setup
- ProtoNet/Siamese/Triplet for few-shot/open-set

17.4 Mandatory ablations

- Full CID-GaitNet + CycleDIR
- minus direction head
- minus bag head
- minus adversarial nuisance heads
- minus orthogonality loss
- minus supervised contrastive loss
- minus ArcFace, CE only
- minus CycleDIR
- amplitude only
- phase only
- amplitude + phase
- no augmentation
- no hard positive mining

17.5 Publication strategy

64. Write Paper 1 first using CID-GaitNet + CycleDIR on the 25-person regular-router subset.
65. Make cross-direction and normal-to-bag robustness the main story; do not rely on random accuracy.
66. Use few-shot/open-set verification to strengthen the biometric framing.
67. Write Paper 2 second using PAIR-BridgeNet for regular-to-modified and modified-to-regular transfer.
68. Frame the modified-router setup as a paired pilot cross-configuration domain shift, not universal antenna superiority.
69. Collect modified-router data for more participants if possible; increasing from 5 to 15-25 would greatly strengthen Paper 2.
70. Release metadata, split files, and code/configs if publication policy allows.
71. Use LLM-inspired methods only as masked/autoregressive CSI pretraining or as a reporting layer, not as a pure LLM classifier.

18. References and source notes

Dataset source notes: [D1] Uploaded document "CSI Dataset Structure.docx". [D2] Uploaded pasted research prompt describing the same dataset and requested Q1-level analysis.

Literature and technical references should be converted to full bibliographic entries before manuscript submission. The URLs below are included as traceable source notes from the research discussion.

- [L1] WiWho: WiFi-based person identification by gait. URL: https://yunzezeng.com/assets/pdfs/WiWho_IPSN_2016.pdf
- [L2] WiFi-ID: Human identification using WiFi signal. URL: https://www.cse.unsw.edu.au/~wenh/zhang_dcoss16.pdf
- [L3] Gate-ID: WiFi CSI direction-invariant identification. URL: https://researchportal.northumbria.ac.uk/ws/files/52102139/FINAL_VERSION.pdf
- [L4] CAUTION: Few-shot open-set WiFi human authentication. URL: https://oar.a-star.edu.sg/storage/d/d6rq0y7ex7/caution.pdf
- [L5] AutoFi: Geometric self-supervised learning for WiFi sensing. URL: https://arxiv.org/abs/2205.01629

- [L6] SenseFi: Benchmark/library for WiFi sensing. URL: https://www.sciencedirect.com/science/article/pii/S2666389923000405
- [L7] Wi-Gait: Multiscale residual convolution-enhanced Transformer for gait identification. URL: https://www.sciencedirect.com/science/article/abs/pii/S1389128623001962
- [L8] DCS-Gait: Cross-domain/channel-state CSI gait recognition. URL: https://xplorestaging.ieee.org/document/10411052
- [L9] MaP-SGAN: Multi-anchor point Siamese GAN for cross-domain CSI gait recognition. URL: https://www.sciencedirect.com/science/article/pii/S0957417424009497
- [L10] NeuralWiGait: CNN-BiGRU for WiFi gait recognition. URL: https://link.springer.com/article/10.1007/s11227-024-06878-0
- [L11] ViT-family WiFi CSI HAR example. URL: https://colab.ws/articles/10.1109/jiot.2024.3375337
- [L12] WiSigPro Transformer WiFi sensing. URL: https://www.sciencedirect.com/science/article/pii/S0957417424018438
- [L13] Multi-view CSI gait fusion. URL: https://www.sciencedirect.com/science/article/pii/S0957417425039594
- [L14] Transformer amplitude/phase stationary person ID. URL: https://arxiv.org/abs/2507.12854
- [L15] CSI-Bench benchmark. URL: https://ai-iot-sensing.github.io/projects/project.html
- [L16] WiFi CSI leakage/splitting critique. URL: https://www.mdpi.com/1424-8220/24/10/3159
- [L17] Phase sanitization and preprocessing reference. URL: https://www.sciencedirect.com/science/article/pii/S2542660523002834
- [L18] PatchTST time-series Transformer. URL: https://github.com/haoxin1998/PatchTST-1
- [L19] Masked time-series autoencoder. URL: https://arxiv.org/abs/2301.08871

Final caution: before journal submission, verify every reference again, replace URLs with formal bibliographic metadata, and update the related-work table with the latest papers from the target journal database.